

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 2

Kopiereg voorbehou

Semester Test 2

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Module EBN122
Elektrisiteit en Elektronika
October 2016

Module EBN122
Electricity and Electronics
October 2016



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Exam information:

Eksamensinligting:

Maximum marks: Maksimum punte:	45	Full marks: Volpunte:	46
Duration of paper: Duur van vraestel:	90 minutes 90 minute	Open / closed book: Oopboek / toeboek:	Closed Toe
Total number of pages (including this page): Totale aantal bladsye (hierdie blad ingesluit):	13		

IMPORTANT - BELANGRIK

- The examination regulations of the University of Pretoria apply.*
Die eksamenregulasies van die Universiteit van Pretoria geld.
- All questions must be answered on the ANSWER SHEET.*
Alle vrae moet op die ANTWOORDBLAD beantwoord word.
- Students may do their own rough calculations on the question paper – the question paper will not be taken in.*
Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
- Answers must include units!*
Antwoorde moet eenhede insluit!
- Final answers must be written as real numbers, and not as fractions.*
Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.

Dosent(e): Lecturer(s):	1. D le Roux
	2. R dos Santos
	3. J Joubert

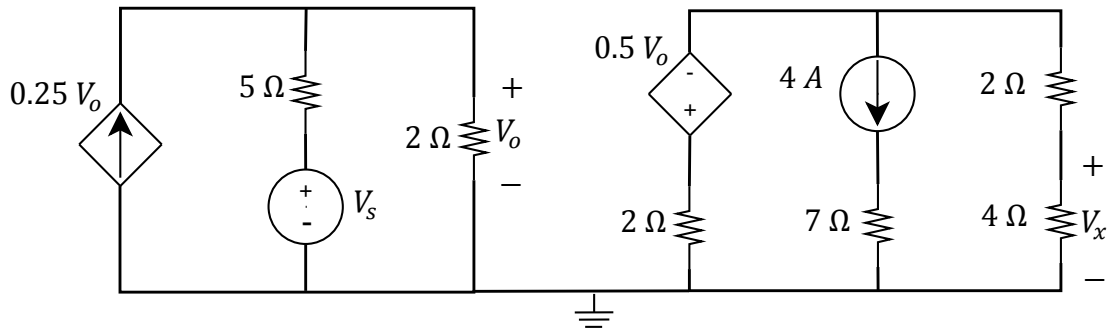
VRAAG/QUESTION 1 [8]

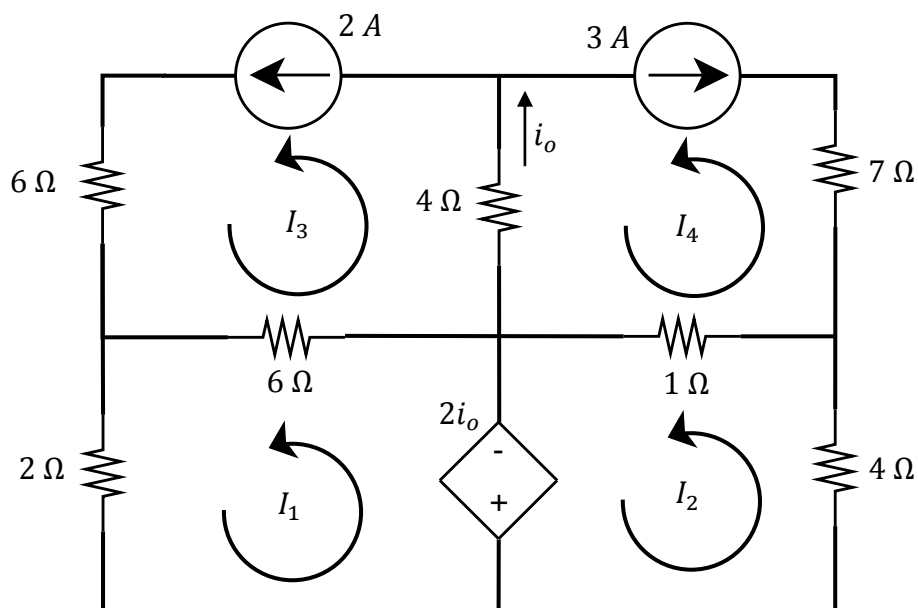
Question 1.1 [4]

Using nodal analysis solve for V_s and V_x , if given that $V_o = 8\text{ V}$.

Vraag 1.1 [4]

Gebruik node analise om V_s en V_x op te los, indien gegee dat $V_o = 8\text{ V}$.



Question 1.2 [4]Using mesh analysis determine I_1 and I_2 .**Vraag 1.2 [4]**Gebruik lusanalyse om I_1 en I_2 op te los.

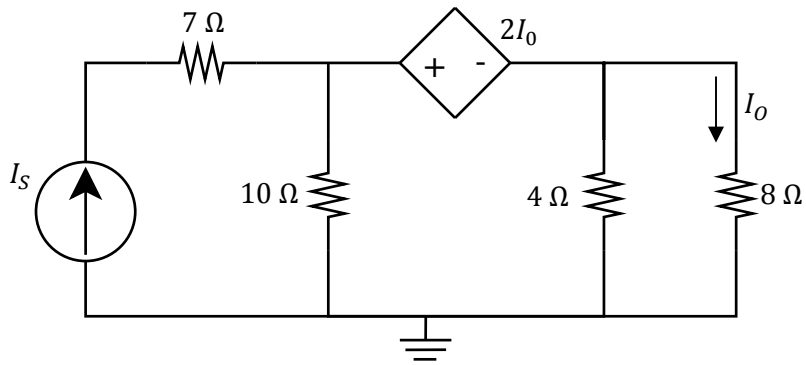
VRAAG/QUESTION 2 [22]

Question 2.1 [2]

Using the principle of linearity, determine the ratio: $\frac{I_0}{I_s}$.

Vraag 2.1 [2]

Deur gebruik te maak van die beginsel van lineariteit, bepaal die verhouding: $\frac{I_0}{I_s}$.

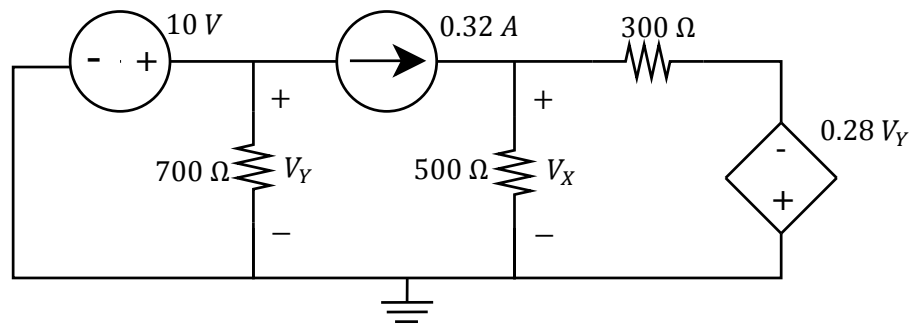


Question 2.2 [4]

- a) Determine the individual contribution of the 320 mA current source to V_X .
- b) Determine the individual contribution of the 10 V voltage source to V_X .

Vraag 2.2 [4]

- a) Bepaal die unieke bydra van die 320 mA stroombron tot V_X .
- b) Bepaal die unieke bydra van die 10 V spanningsbron tot V_X .

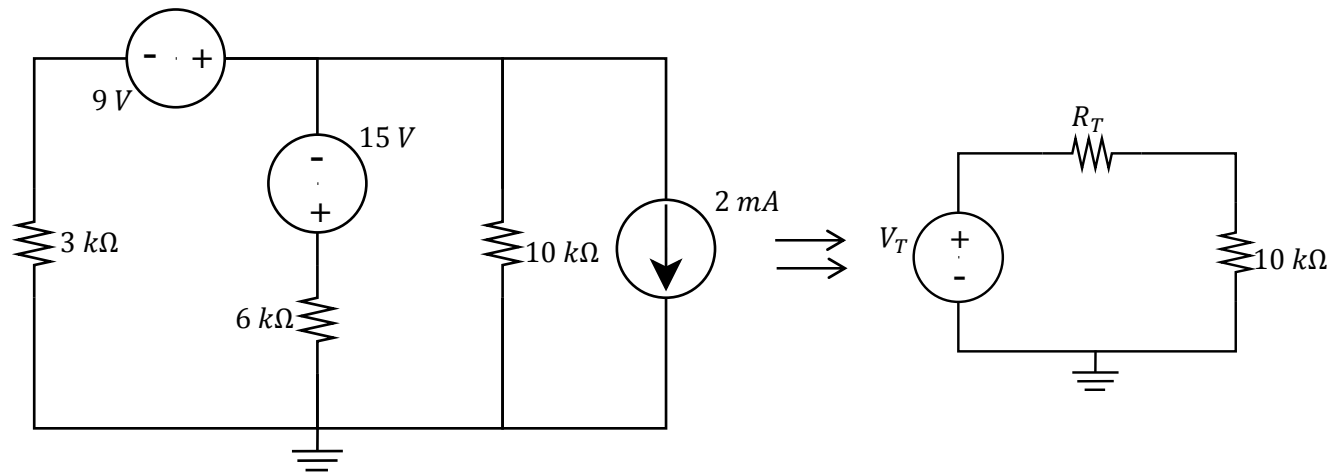


Question 2.3 [4]

The aim is to simplify the circuit on the left to obtain the circuit on the right with respect to the $10\text{ k}\Omega$ resistor. Make use of source transformation and determine V_T and R_T .

Vraag 2.3 [4]

Die doel is om die linkerkantse kringbaan te vereenvoudig na die regterkantse kringbaan met verwysing tot die $10\text{ k}\Omega$ weerstand. Maak gebruik van brontransformasie en bepaal V_T en R_T .

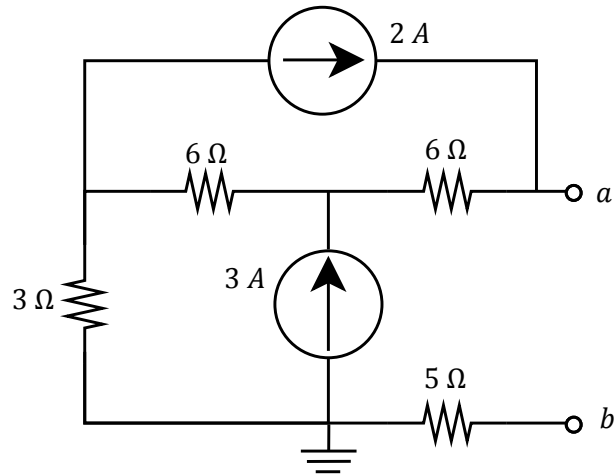


Question 2.4 [4]

Determine the Thevenin equivalent voltage V_{TH} and resistance R_{TH} w.r.t. terminals a - b .

Vraag 2.4 [4]

Bepaal die Thevenin ekwivalente spanning V_{TH} en weerstand R_{TH} t.o.v. terminale a - b .

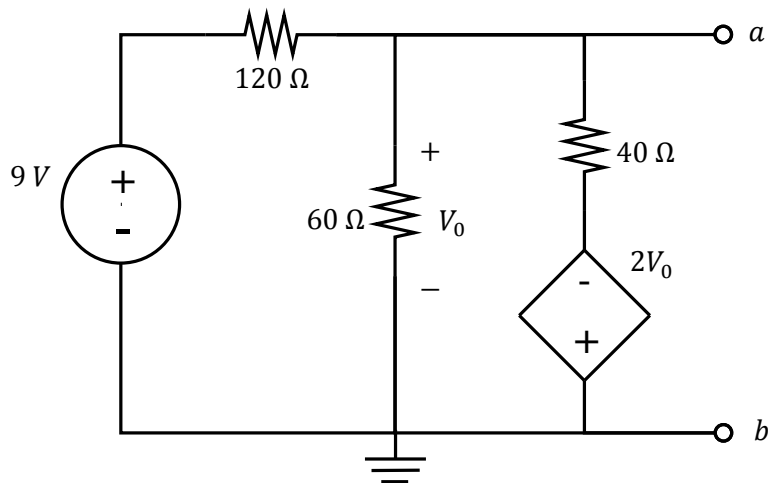


Question 2.5 [4]

Determine the Norton equivalent current I_N and resistance R_N wrt terminals a - b .

Vraag 2.5 [4]

Bepaal die Norton ekwivalente stroom I_N en weerstand R_N tov terminale a - b .



Question 2.6 [4]

The short circuit current of a circuit is $I_N = 6 \text{ mA}$. The circuit delivers maximum power to a load of $R_L = 10 \text{ k}\Omega$.

- a) What is the maximum power P_{max} absorbed by the load resistor R_L ?
- b) If a resistor of $15 \text{ k}\Omega$ is placed in parallel to the load resistor R_L , what is the total power absorbed by the parallel combination? (I.e. the total power absorbed by the combined resistance ($15 \parallel 10$) $\text{k}\Omega$.)

Vraag 2.6 [4]

Die kortsluitstroom van 'n kringbaan is $I_N = 6 \text{ mA}$. Die kringbaan lewer maksimum drywing aan 'n las van $R_L = 10 \text{ k}\Omega$.

- a) Wat is die maksimum drywing P_{max} wat deur die lasweerstand verkwis word?
- b) Indien 'n weerstand van $15 \text{ k}\Omega$ parallel aan die lasweerstand R_L gekoppel word, wat is die totale drywing wat deur die parallele kombinasie geabsorbeer word? (M.a.w. die total drywing geabsorbeer deur die gekombineerde weerstand ($15 \parallel 10$) $\text{k}\Omega$.)

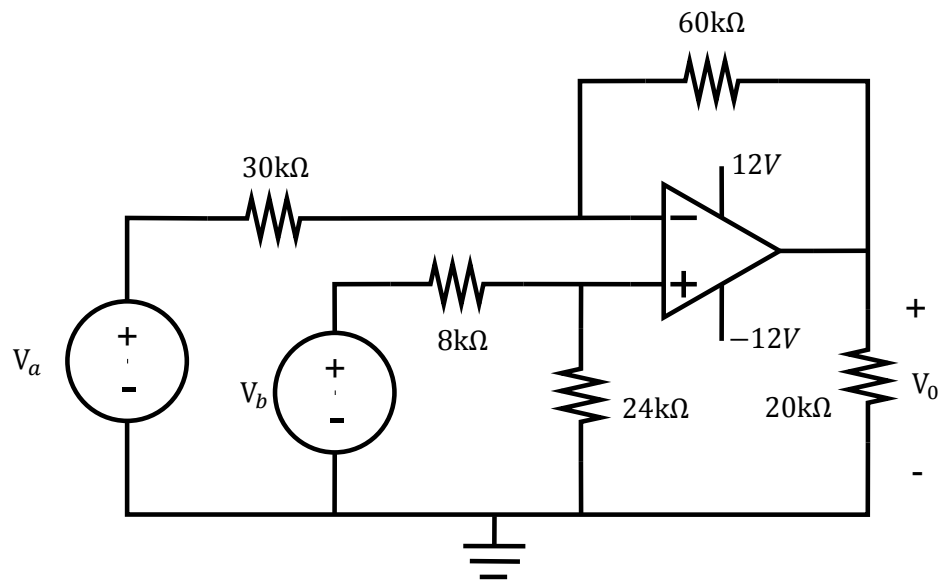
VRAAG/QUESTION 3 [16]

Question 3.1 [4]

- a) Calculate V_0 if $V_a = 2V$ and $V_b = 1V$.
b) Calculate V_0 if $V_a = -3V$ and $V_b = 0V$.

Vraag 3.1 [4]

- a) Bereken V_0 as $V_a = 2V$ en $V_b = 1V$.
b) Bereken V_0 as $V_a = -3V$ en $V_b = 0V$.

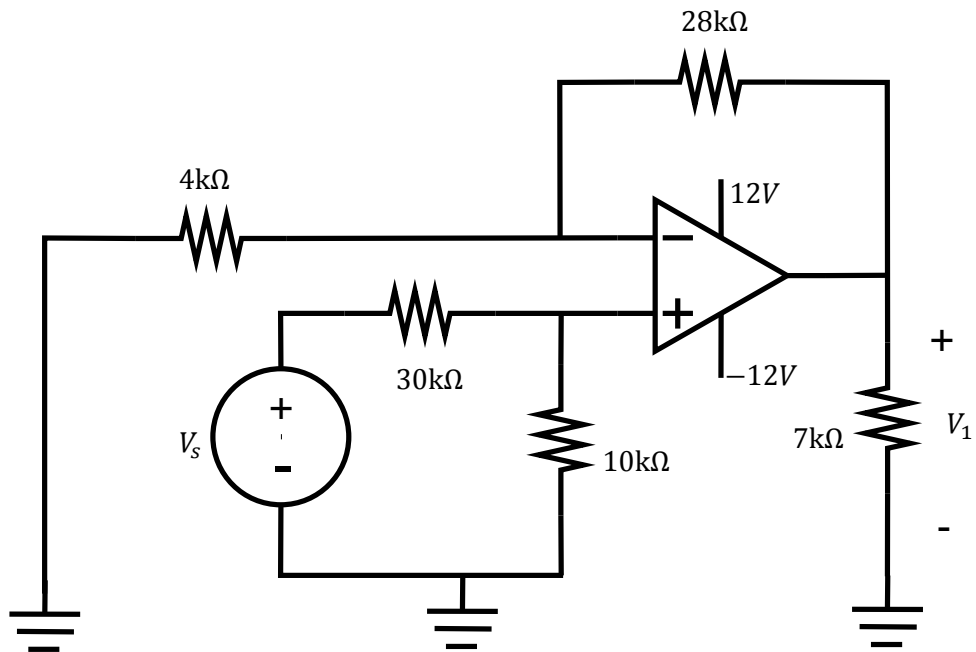


Question 3.2 [4]

- a) Determine V_1 if $V_s = 4V$.
b) Determine the range of values of V_s that will ensure that the op-amp operates in the linear region.

Vraag 3.2 [4]

- a) Bepaal V_1 as $V_s = 4V$.
b) Bepaal die omvang van waardes van V_s wat sal verseker dat die operasionele versterker in sy lineêre gebied sal funksioneer.

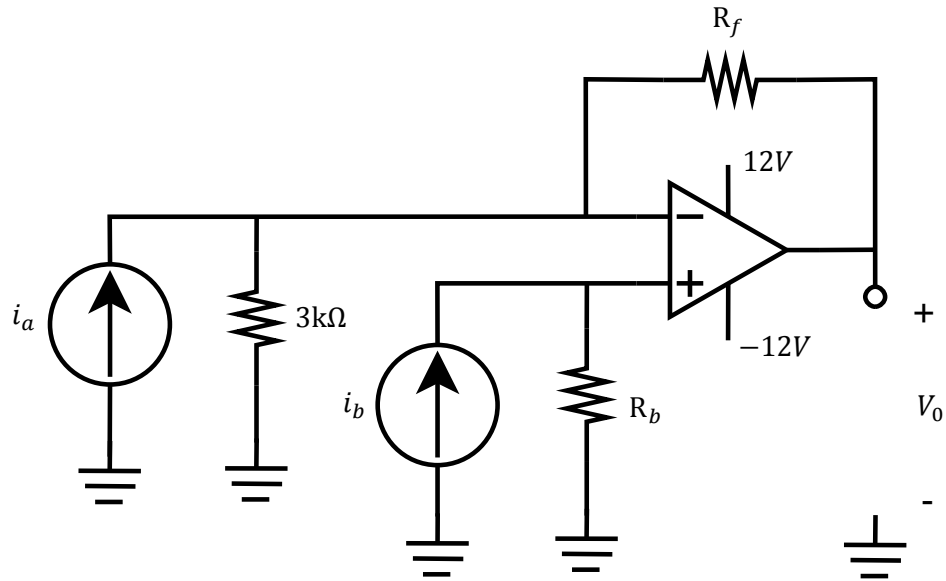


Question 3.3 [4]

Determine values of R_b and R_f so that
 $V_0 = 2000i_b - 2000i_a$.

Vraag 3.3 [4]

Bepaal waardes vir R_b en R_f sodat
 $V_0 = 2000i_b - 2000i_a$.



Question 3.4 [4]

- a) Determine the amplification factor V_2/V_1 .
b) Determine i_3 if $V_2 = 2V$.

Vraag 3.4 [4]

- a) Bepaal die versterkingsfaktor V_2/V_1 .
b) Bepaal i_3 as $V_2 = 2V$.

